REDCap advanced features

Quick reference: field embedding, piping, action tags, smart variables, special functions

Supplementary material for REDCap Part 2, Research Data Management, MEPI, University of Antwerp, 2026-27. Examples come from TBCARE (class demo) and PISTIC (homework). The full, current help pages are inside REDCap: in the Online Designer, click Action Tags, Piping, Smart Variables, Field Embedding or Special Functions.

1 Field embedding

Moves a field to a place you choose on the same form, for example into a table. Write the variable name in **curly brackets** in the label, field note, section header or choice label of another field.

You write (label of a descriptive field)	Result
`<table><tr><th>Weight</th><th>Height</th></tr> <tr><td>{weight_kg}</td><td>{height_cm}</td></tr></table>`	The two input boxes appear side by side in a table.
`I agree to take part {cons_part}`	The Yes / No buttons appear after the sentence (TBCARE e-Consent).
`{known_time} {known_unit}`	Number and unit on one line (PISTIC FSX item 1g).

- Same instrument only; on a multi-page survey, the same page.
- The embedded field keeps its validation, required status and branching logic. Its own label is not shown.
- Embed a field only once. Test it on a real record: the preview in the Online Designer is only approximate.

2 Piping

Shows an answer that was already given, inside new text. Write the variable name in **square brackets**.

You write	REDCap shows
`Of these [sex_times] times, how many with a condom?`	Of these 6 times, how many with a condom?
`[sex]` and `[sex:value]`	The label (Female) and the code (2) of a multiple-choice answer
`[contraception:checked]`	The ticked options of a checkbox field
`[baseline_arm_1][weight_kg]`	The value at a named event
`[first-event-name][first_name]`	The value at the first event of the record's arm (works in every arm)
`[ae_term]` in a repeating form	The value of the same instance

Piping works in labels, field notes, section headers, choice labels, survey instructions and completion text, survey invitations, alerts, custom record labels, and the `@DEFAULT` action tag.

3 Action tags

Write them in the **Field Annotation** of a field. Several can be combined, separated by a space.

Action tag	What it does	Example
`@TODAY @NOW`	Fills in today's date or the date and time	Visit date
`@HIDDEN @HIDDEN-SURVEY @HIDDEN-FORM`	Hides the field everywhere, on surveys only, or on forms only	A date staff need, participants not
`@READONLY`	Shows the field but it cannot be changed	A calculated text
`@CHARLIMIT=n`	At most n characters, with a counter	Initials, phone numbers
`@CALCTEXT(…)`	A calculation that returns text	`@CALCTEXT(if([eligible] = 1, 'ELIGIBLE', 'NOT ELIGIBLE'))`
`@CALCDATE(…)`	A date calculated from another date	Next visit 91 days after Day 0
`@NONEOFTHEABOVE='6'`	Choosing code 6 unticks all other options	Symptoms: None
`@HIDECHOICE='3'`	Hides an option but keeps data already entered with it	An answer option you retired
`@DEFAULT='…'`	A starting value; can pipe	`@DEFAULT='[record-dag-name]'`
`@LATITUDE @LONGITUDE`	Captures the GPS position of the device	Home position (SDE section A)
`@WORDLIMIT=n`	At most n words	Free-text comments

4 Smart variables

Filled in by REDCap from the context. Use them in logic, calculations and piping.

Smart variable	Gives
`[record-name]`	The record ID
`[record-dag-name] [record-dag-label]`	The record's Data Access Group (unique name or label)
`[event-name] [event-label]`	The current event (unique name or label)
`[first-event-name] [previous-event-name] [next-event-name]`	Events relative to the current one, within the record's arm
`[arm-number] [arm-label]`	The arm of the record
`[user-name] [user-dag-name]`	The user who is logged in and their DAG
`[current-instance]`	The instance number in a repeating form or event
`[survey-link:fsx]`	A link to a survey

Example from PISTIC: hide the swab questions of FGE in arm 3 with `[gen_exam_done] = '1' and [arm-number] <> '3'`.

5 Special functions

Function	Example
`datediff(d1, d2, unit)`	`rounddown(datediff([dob], [scr_date], 'y'), 0)`: age in whole years
`if(condition, yes, no)`	`if([qtcf] > 500, 1, 0)`
`round() rounddown() roundup()`	`round([weight_kg] / ([height_cm] / 100) ^ 2, 1)`: BMI
`sum() mean() min() max()`	`sum([qol_1], [qol_2], [qol_3], [qol_4])` (blanks are ignored)
`isblankormissingcode()`	`isblankormissingcode([dob])`: true if empty
`concat() left() upper()`	`upper(left([last_name], 2))`
`contains() starts_with()`	`contains([ae_term], 'rash')`

Before you trust a calculation, make a test record where you know the answer, and check what happens when a value is missing.